National University of Computer and Emerging Sciences

Lahore Campus

Electromagnetic Theory (EE3005)

Sessional-II Exam

Total Time (Hrs): 1
Total Marks: 40
Total Questions: 2

Date: October 31, 2026

Course Instructor(s)

1. Dr. Afzal Ahmed (Course Moderator)
2. Mohsin Yousuf

[redacted] Roll No [redacted] Section [redacted] Student Signature

Do not write below this line. Question paper is exam material, you're not allowed to take it along.

1. *Attempt all questions and remember to solve parts of the same question together.*
2. *Final answers should be correct up to two decimal places with proper SI units.*
3. *Show all the steps with the help of diagrams and equations.*

CLO # 02: Analyze** electrostatic fields and/or its properties governed by Coulomb's / Gauss's law for a given charge distribution in free space and / or dielectrics. **[C4]

Q1: **[20]**

(a) State Divergence Theorem and **deduce** Maxwell's first equation from Gauss's law and **identify** the relation between them. **(5)**

(b) An electric flux density vector field inside an unconventional spherical sector is modeled by: **(15)**

$$\mathbf{D} = \frac{12}{r}\cos(2\theta)\,\boldsymbol{a}_\theta \quad \left[\frac{C}{m^2}\right]$$

The active field system is structurally confined within the geometric space co-ordinates:

$$1 < r < 2\,m, \qquad 0 < \theta < \frac{\pi}{2}\,rad, \qquad 0 < \phi < \frac{\pi}{2}\,rad$$

Analyze this electromagnetic environment using two structurally distinct analytical frameworks to **illustrate** the net accumulated system charge Q_{encl}. Your analysis must explicitly break down the following structural layers:

i. **Deconstruct** Boundary Face Contributions using L.H.S. of the Divergence Theorem (*Method 1*).

ii. **Isolate** Spatial Divergence Components using the R.H.S. of the Divergence Theorem (*Method 2*).

iii. **Contrast** and **Verify** Frameworks: Analyze why the mathematical canceling behaviors in both separate methods converge perfectly on an identical scalar conclusion.

$$\text{div}\,\mathbf{D} = \frac{1}{r^2}\frac{\partial}{\partial r}(r^2 D_r) + \frac{1}{r\sin\theta}\frac{\partial}{\partial \theta}(\sin\theta\, D_\theta) + \frac{1}{r\sin\theta}\frac{\partial D_\phi}{\partial \phi}$$

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CLO # 02: Analyze** electrostatic fields and/or its properties governed by Coulomb's / Gauss's law for a given charge distribution in free space and / or dielectrics. **[C4]

Q2: **[20]**

(a) Consider the scalar electric potential field in a three-dimensional region of space described by $V(x, y, z) = 4x^2 - 3y^2 + 2z^2 - 4xz$ [V] and let $P(3, -2, 1)$ be a point in this region. **Calculate** the following at P with clear justification and proper mathematical steps: **(8)**

- **i.** The electric potential V_P.
- **ii.** The electric field intensity $\mathbf{E}_P$ and its direction.
- **iii.** The electric flux density $\mathbf{D}_P$.
- **iv.** The volume charge density ρ_v.

~~$V = QE$~~ ~~$W =$~~

$V = \frac{Q}{4\pi\varepsilon_0 R}$

(b) Four out of five identical point charges, each of 1.2 [nC], are positioned at the four corners of a square of side length 5 [cm] in free space and fifth charge is installed at the center of the square. Draw the constellation and **find** the total electric potential energy W_E stored in the configuration. **(12)**

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